

AIM: C1 implements C2 extends I1.

Steps for execution:

1. Open Eclipse.
2. Create a Java Project.
3. Create a package named record.
4. Create a class named C1ImplC2ExtI1.
5. Enter/paste the complete code.
6. Save the program using Ctrl + S.
7. Right-click on the program.
8. Select Run As → Java Application.
9. The program first displays the message from add().
10. Then it asks you to enter the value of r.
11. Enter a number.
12. The program displays $r + 20$.
13. Output:

 this is implements from interface

 enter the value of r :

 20

 the sum of two values is :40

Algorithm:

1. Start.
2. Create interface I4.
3. Declare constant y and method add() in I4.
4. Create class Demo.
5. Create a Scanner object to take input.
6. Define the diff() method in Demo.
7. Read the value of r.
8. Calculate $r + 20$ and display the result.
9. Create class C1ImplC2ExtI1.
10. Extend the Demo class.
11. Implement the I4 interface.

12. Define the add() method.
13. Create an object re of C1ImplC2Ext1.
14. Call add().
15. Call the inherited diff() method.
16. Display the results.
17. Stop.

Program:

```
package record;

import java.util.Scanner;

// I4 is an interface
interface I4 {

    // y is a constant variable in the interface
    int y = 45;

    // Abstract method declared in the interface
    void add();
}

// Demo is a class
class Demo {

    // Creating Scanner object to take input from the user
    Scanner sc = new Scanner(System.in);

    // Variable to store the input value
    int r;

    // Method to calculate the sum
    void diff() {

        // Asking the user to enter a value
        System.out.println("enter the value of r :");

        // Reading the value entered by the user
        r = sc.nextInt();

        // Adding r with 20 and displaying the result
        System.out.println("the sum of two values is : " + (r + 20));
    }
}
```

```

}

// C1ImplC2Extl1 extends Demo and implements I4

// Therefore, it inherits diff() from Demo

// and must implement add() from I4

public class C1ImplC2Extl1 extends Demo implements I4 {

@Override

    // Implementation of add() method from I4

    public void add() {

        System.out.println("this is implements from interface");

    }

// Main method - execution starts here

    public static void main(String[] args) {

// Creating an object of C1ImplC2Extl1

        C1ImplC2Extl1 re = new C1ImplC2Extl1();

// Calling add() method

        re.add();

// Calling diff() method inherited from Demo

        re.diff();

    }

}

```

Output:

this is implements from interface

enter the value of r :

20

the sum of two values is :40

Output Screenshot:

```
Console
-terminated> C:\mpic2\src\1\ [Java Application] C:\Program Files\Java\jdk-25.0.3\bin\java.exe (29-Aug-2025, 9:38:43 pm - 9:38:47 pm elapsed 0:00:04) [pid: 7948]
this is implements from Interface
enter the value of n :
20
the sum of two values is :40
```